3D PERCEPTION · PROJECT 2

Classical camera–LiDAR fusion on KITTI

KITTI LiDAR and Camera classical fusion techniques comparison

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OBJECTIVE

Does a camera help a classical LiDAR detector?

01

Fuse

Camera and LiDAR at the early, mid and late level.

02

Without learning

Only geometry, probabilities and hand-set rules.

03

Measure

Same frames, same KITTI protocol, each sensor alone as reference.

METHODS

L1 LiDAR only	**C1** camera only	**FE** early fusion	**FM** mid fusion	**FL-c** late fusion	**L2** learned contrast

RECAP · PROJECT 1

Same LiDAR pipeline as project 1

01
Ground removal
RANSAC plane

02
BEV grid
log-polar 200 × 1000, 2–64 m

03
Clustering
8-connected, 3 × 3 closing

04
Box fitting
oriented, by edge distance

05
Classify
size rules, cars extended to the KITTI mean

06
Evaluation
3D IoU against the KITTI labels

L1 ON VAL · BIG NUMBER = F1

Car P 0.23 · R 0.43	**Pedestrian**	**Cyclist**
0.301	**0.014**	**0.053**

Same detections per frame as project 1, now on the standard val split so every method sees the same 3769 frames.

DATA

LiDAR plus the colour stereo pair

3769

val frames for every result; 3712 train frames only for tuning and training

0.54 m

stereo baseline between the two cameras

2

rectified colour cameras, image_2 and image_3, 1242 × 375

~330k

stereo points vs ~120k LiDAR points per frame

Every parameter chosen on **train**, every number on **val**.

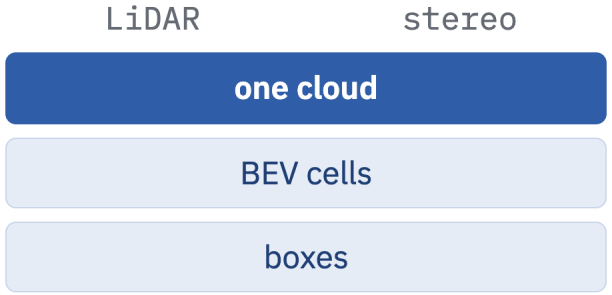
FUSION · LEVELS

Three places where the sensors meet

01 · FE

Early

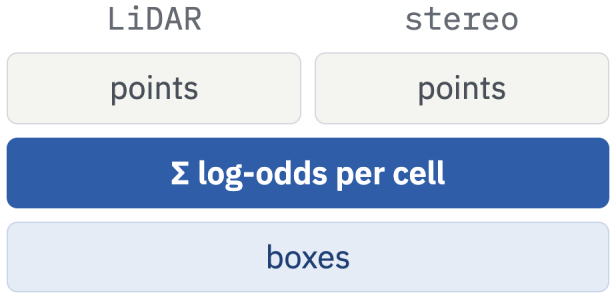
Raw points: the two clouds become one.



02 · FM

Mid

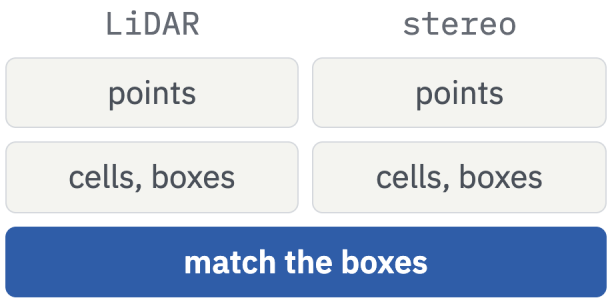
Intermediate representation: evidence per BEV cell.



03 · FL-c

Late

Final decisions: two detection lists.



Same inputs for all three: the **L1** LiDAR pipeline and the **C1** stereo cloud.

CAMERA · 01

Depth from a rectified pair

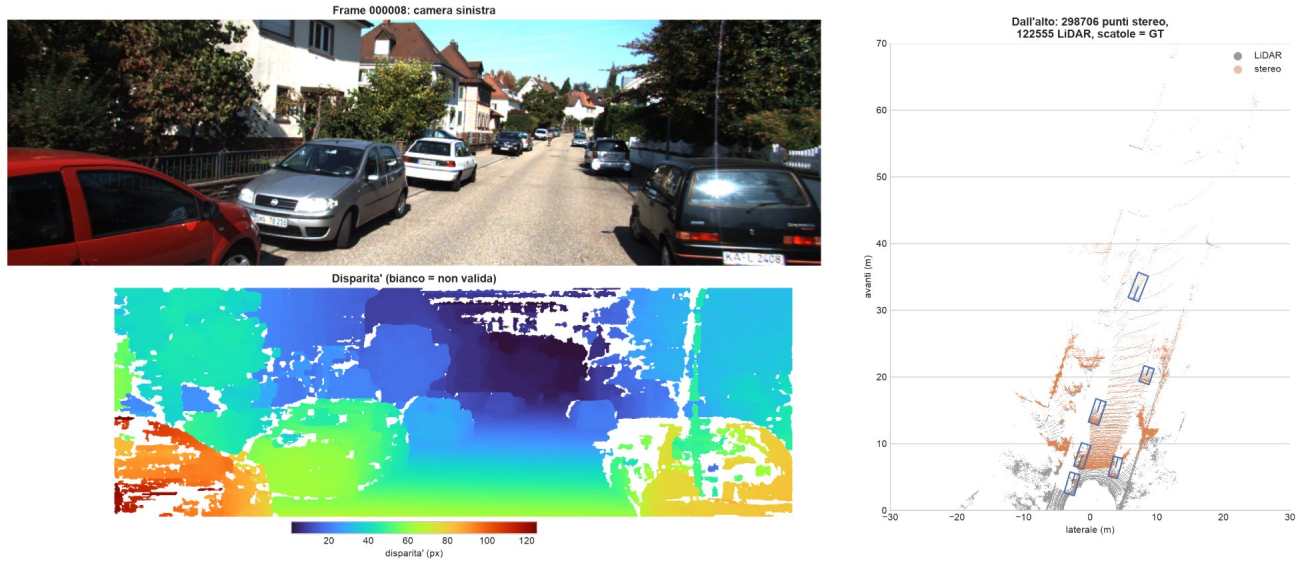
OpenCV SGBM on the grayscale pair

disparities	128
block	7 × 7
smoothness P1 · P2	$8 \cdot 7^2$ · $32 \cdot 7^2$
uniqueness	10
speckle filter	on
left-right check	1 px

SIMILAR TRIANGLES

$$Z = f \cdot B / d$$

f focal, **B** baseline from P2 and P3 (B ≈ 0.54 m), **d** disparity in pixels



Frame 8: left image, disparity (white = invalid), and from above the stereo points (orange) over the LiDAR (grey), boxes = labels.

CAMERA · 02

Stereo noise grows with distance²

$$\Delta Z \approx Z^2 / (f \cdot B) \cdot \Delta d$$

depth error for a matching error of Δd = 0.25 px

at 5 m	at 20 m	at 40 m
0.02 m	**0.26 m**	**1.0 m**

C1 SETTINGS

01

Flying pixels removed

depth jump > 5% in the 3 × 3 neighbourhood

02

Height and range band

only points 0.3–2.5 m above ground and ≤ 40 m

03

Dense cells only

a BEV cell is occupied only with ≥ 25 points

Median depth error vs LiDAR on frame 8: **0.17 m (1.7%)**.

FUSION · EARLY · FE

One cloud

LiDAR points + stereo points beyond 30 m → **one cloud** → L1 pipeline → boxes

TRIED · ALL STEREO POINTS

Adding every stereo point breaks the LiDAR clusters

0.25 → 0.04

Car F1 on train, L1 alone vs LiDAR plus the whole stereo cloud

KEPT · FAR STEREO ONLY

Stereo only where the LiDAR rings get sparse

> 30 m

stereo points added only beyond 30 m, one point per 20 cm cube to match the LiDAR density

FUSION · MID · FM

Bayes on the BEV grid

LiDAR grid + stereo grid → **Σ log-odds** → L > 0.3 → boxes from LiDAR points

LOG-ODDS PER CELL

$$L = \Sigma \log(p_s / (1 - p_s))$$

independent sensors s and a 0.5 prior: Bayes becomes a sum

Occupied if L > 0.3, clustered as in L1; boxes fitted on the LiDAR points only.

`LiDAR · supports`

each object point in a cell adds log(0.6/0.4), up to 3 points

`Stereo · only contradicts`

a cell where it sees ground and no object gets log(0.4/0.6): it cancels one LiDAR point

Positive stereo evidence merged neighbouring objects, so it was dropped.

FUSION · LATE · FL-C

Keep what both sensors see

L1 detections + C1 detections → **Hungarian association** → confirmed LiDAR detections

GATES · BEV DISTANCE BETWEEN BOX CENTRES

2 m	1 m	1.5 m
Car	Pedestrian	Cyclist

Kept

LiDAR detections with a camera match; **box and class from the LiDAR**, whose geometry is far more accurate.

Tried

Keeping single-sensor detections with a lower confidence: worse on train.

EVALUATION

Same protocol as project 1, two additions

Same as project 1

Boxes compared in the LiDAR frame

Greedy matching by decreasing 3D IoU: 0.7 Car, 0.5 Pedestrian and Cyclist

Moderate subset; Van and Person_sitting ignored

Added · 01

BEV IoU

The same matching, with the overlap measured from above.

Added · 02

Detection confusion matrix

An extra “background” row and “missed” column.

	Car	Ped	Cyc	missed
Car	TP	swap	swap	FN
Ped	swap	TP	swap	FN
Cyc	swap	swap	TP	FN
background	FP	FP	FP	–

Rows = labels, columns = detections.

RESULTS · 3769 VAL FRAMES · 3D IOU · MODERATE

Late fusion is the only classical fusion that helps

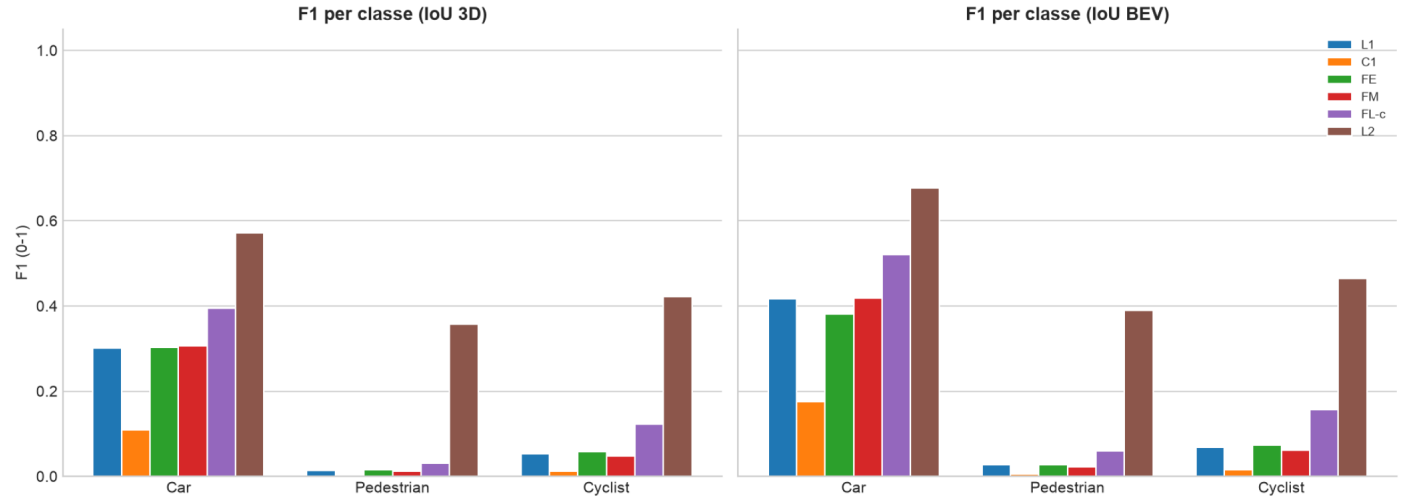
Method	Car TP	Car FP	Car P	Car R	Car F1	Ped F1	Cyc F1
L1 LiDAR	3375	11179	0.232	0.429	0.301	0.014	0.053
C1 stereo	1246	13719	0.083	0.158	0.109	0.003	0.012
FE early	3030	9066	0.250	0.385	0.303	0.015	0.058
FM mid	3444	11129	0.236	0.437	0.307	0.012	0.048
FL-c **late**	3099	4747	0.395	0.394	**0.394**	0.030	0.123

CONTRAST · LEARNED DECISION, NOT A FUSION

L2 PointNet	3841	1736	0.689	0.488	**0.571**	**0.358**	**0.423**

RESULTS · F1 · 3D AND BEV IOU

Late fusion lifts all three classes



Same picture with BEV IoU: late fusion is the best classical method in every class; L2, the learned contrast, is above all.

CAR F1 VS L1

Early
FE
+0.002

Mid
FM
+0.006

Late
FL-c
+0.093

Late fusion
car false positives
−58%

RESULTS · QUALITATIVE

Same frame, six methods

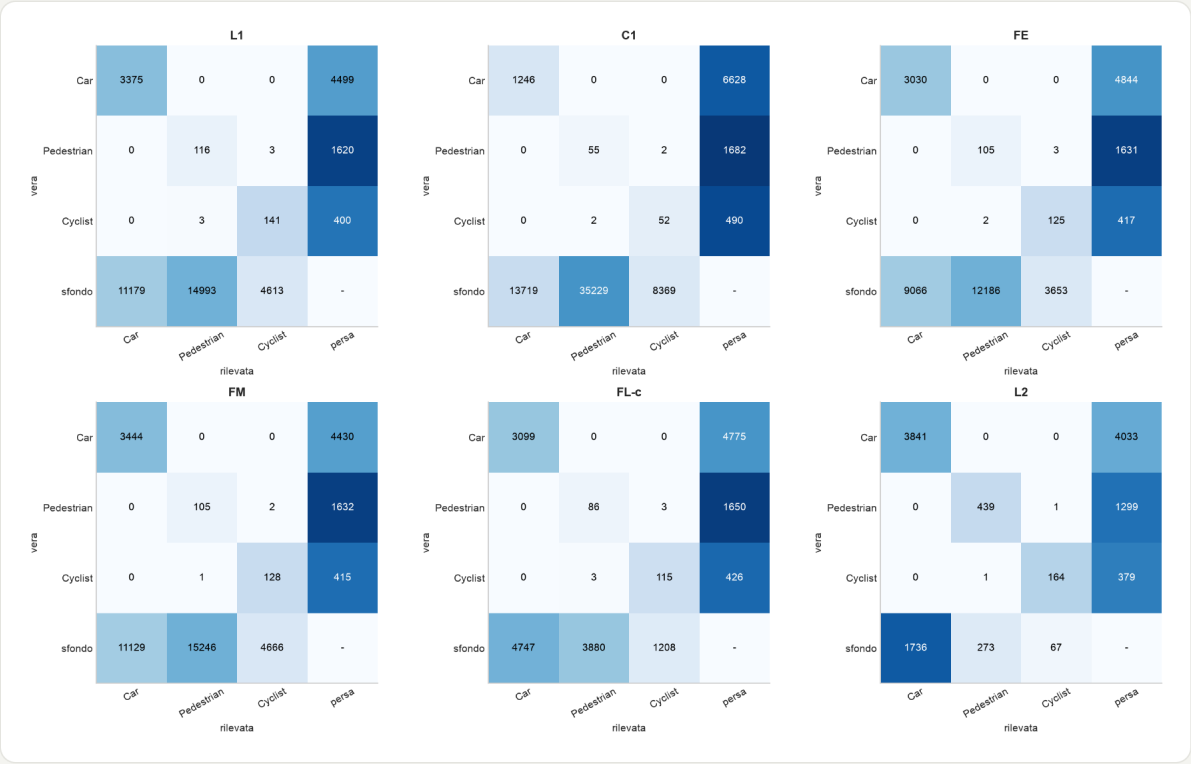


green · correct (TP) red · invented (FP) yellow · missed (FN) grey · on an ignored label

L1 23 false positives → FL-c 10 → L2 0, the car found by all LiDAR methods, missed by C1.

RESULTS · ERRORS

Wrong place, not wrong class



Almost no class swaps

in any method: the Car, Pedestrian and Cyclist cells off the diagonal stay near zero.

Missed and invented objects

the errors sit in the “missed” column and in the “background” row.

Late fusion empties the background row

car false positives 11179 → 4747: only what the camera confirms survives.

`vera = label · rilevata = detection`
`sfondo = background · persa = missed`

CONTRAST · L2

What a small network changes

Same clusters as L1, only the decision is learned: a small PointNet.

L1 cluster points → **shared MLP 3→64→128→256** → max pooling → + box size, range, point count → class, incl. background; 7-value box correction

~100k labelled clusters from train · 20 epochs · ~15 min on a laptop GPU

0.57
Car F1, from 0.30

0.36
Pedestrian F1, from 0.01

0.42
Cyclist F1, from 0.05

1736
car false positives, from 11179

The bottleneck of the classical pipeline is **the rule-based decision**, not the missing sensor.

CONCLUSIONS

What fusion buys a classical detector

Works

Late fusion: Car F1 +31%, −58% false positives

Every step readable

No training

Breaks

Stereo depth noise grows with Z^2

So raw points and cells (early, mid) barely help

Camera alone far behind at IoU 0.7

Next

Learned decision on fused data

AP with confidence scores

Better stereo matching

> **With a noisy second sensor, fusing decisions is more robust than fusing data.**

THANK YOU

Questions?

Maximiliano Mamone



BACKUP · QUALITATIVE

A second frame, same pattern



green · correct (TP) red · invented (FP) yellow · missed (FN) grey · on an ignored label

L1 9 false positives → FL-c 4 → L2 1; every LiDAR method finds 2 cars and misses 2.